

**Material Safety Data Sheet****SENSAT™**

MSDS Number: 102000012503

MSDS Version 1.2

Revision Date: 02/22/2012

Print Date: 02/24/2017

**SECTION 1. CHEMICAL PRODUCT AND COMPANY INFORMATION**

|                             |                             |
|-----------------------------|-----------------------------|
| <b>Product name</b>         | SENSAT™                     |
| <b>MSDS Number</b>          | 102000012503                |
| <b>Product code (UVP)</b>   | 06105067                    |
| <b>EPA Registration No.</b> | 264-995                     |
| <b>Product Use</b>          | Seed treatment, Insecticide |

Bayer CropScience  
2 T.W. Alexander Drive  
Research Triangle PK, NC 27709  
USA

For MEDICAL, TRANSPORTATION or other EMERGENCY call: 1-800-334-7577 (24 hours/day)

**SECTION 2. HAZARDS IDENTIFICATION**

*NOTE: Please refer to Section 11 for detailed toxicological information.*

**Emergency Overview** Caution! Moderate eye irritation. May cause skin irritation. May cause irritation of respiratory tract. Avoid inhalation of vapour or mist. Avoid contact with skin, eyes and clothing.

**Physical State** liquid

**Odor** characteristic

**Appearance** brown

**Exposure routes** Ingestion, Inhalation, Eye contact, Skin contact

**Immediate Effects**

**Eye** Moderate eye irritation. Avoid contact with eyes.

**skin** May cause skin irritation. Avoid contact with skin and clothing.

**Ingestion** May be harmful if swallowed. Do not take internally.

**Inhalation** May cause respiratory tract irritation. Avoid inhalation of vapour or mist.

**Chronic or Delayed Long-Term** This product or its components may have target organ effects.

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**Potential Environmental Effect**      Toxic to fish. Highly toxic to aquatic invertebrates.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| <u>Hazardous Component Name</u>            | <u>CAS-No.</u> | <u>Average % by Weight</u> |
|--------------------------------------------|----------------|----------------------------|
| Spinosad, mixture of Factor A and Factor D | 168316-95-8    | 8.66                       |
| Spinosad Factor A                          | 131929-60-7    |                            |
| Spinosad Factor D                          | 131929-63-0    |                            |
| 1,2-Propanediol                            | 57-55-6        | 10.75                      |

**SECTION 4. FIRST AID MEASURES**

|                                     |                                                                                                                                                                                                                                                                                               |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General</b>                      | When possible, have the product container or label with you when calling a poison control center or doctor or going for treatment.                                                                                                                                                            |
| <b>Eye</b>                          | Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. Call a physician or poison control center immediately.                                                                       |
| <b>skin</b>                         | Take off contaminated clothing and shoes immediately. Wash off immediately with plenty of water for at least 15 minutes. Call a physician or poison control center immediately.                                                                                                               |
| <b>Ingestion</b>                    | Call a physician or poison control center immediately. Rinse out mouth and give water in small sips to drink. DO NOT induce vomiting unless directed to do so by a physician or poison control center. Never give anything by mouth to an unconscious person. Do not leave victim unattended. |
| <b>Inhalation</b>                   | Move to fresh air. If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a physician or poison control center immediately.                                                                                      |
| <b>Notes to physician Treatment</b> | Appropriate supportive and symptomatic treatment as indicated by the patient's condition is recommended. There is no specific antidote.                                                                                                                                                       |

**SECTION 5. FIRE FIGHTING MEASURES**

|                                 |                   |
|---------------------------------|-------------------|
| <b>Flash point</b>              | > 100 °C / 212 °F |
| <b>Autoignition temperature</b> | no data available |
| <b>Lower Flammability Limit</b> | no data available |



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|                                     |                                                                                                                                                                                                                                                                                                        |
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| <b>Upper Flammability Limit</b>     | no data available                                                                                                                                                                                                                                                                                      |
| <b>Explosiveness</b>                | no data available                                                                                                                                                                                                                                                                                      |
| <b>Fire and Explosion Hazards</b>   | In the event of fire the following may be released:<br>Carbon monoxide (CO)<br>Carbon dioxide (CO <sub>2</sub> )                                                                                                                                                                                       |
| <b>Suitable extinguishing media</b> | Foam, Carbon dioxide (CO <sub>2</sub> ), Dry chemical, Water spray                                                                                                                                                                                                                                     |
| <b>Fire Fighting Instructions</b>   | Fight fire from upwind position. Keep out of smoke. Cool closed containers exposed to fire with water spray. Do not allow run-off from fire fighting to enter drains or water courses.<br><br>Firefighters should wear NIOSH approved self-contained breathing apparatus and full protective clothing. |

## SECTION 6. ACCIDENTAL RELEASE MEASURES

|                                |                                                                                                                                                                                                                                                 |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b>    | Keep unauthorized people away. Isolate hazard area. Avoid contact with spilled product or contaminated surfaces.                                                                                                                                |
| <b>Methods for cleaning up</b> | Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Clean contaminated floors and objects thoroughly, observing environmental regulations. |
| <b>Additional advice</b>       | Use personal protective equipment.                                                                                                                                                                                                              |

## SECTION 7. HANDLING AND STORAGE

|                                 |                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling procedures</b>      | Handle and open container in a manner as to prevent spillage. Maintain exposure levels below the exposure limit through the use of general and local exhaust ventilation.                                                                                                                                                                      |
| <b>Storing Procedures</b>       | Store in original container and out of the reach of children, preferably in a locked storage area. Protect from freezing.                                                                                                                                                                                                                      |
| <b>Work/Hygienic Procedures</b> | Wash hands thoroughly with soap and water after handling and before eating, drinking, chewing gum, using tobacco, using the toilet or applying cosmetics.<br><br>Before removing gloves clean them with soap and water. Remove soiled clothing immediately and clean thoroughly before using again. Wash thoroughly and put on clean clothing. |



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## SECTION 8. EXPOSURE CONTROLS / PERSONAL PROTECTION

|                               |                                                                                                                                                                                              |        |        |                      |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|----------------------|
| <b>General Protection</b>     | Follow all label instructions. Train employees in safe use of the product.                                                                                                                   |        |        |                      |
|                               | Follow manufacturer's instructions for cleaning/maintaining PPE. If no such instructions for washables, use detergent and warm/tepid water. Keep and wash PPE separately from other laundry. |        |        |                      |
| <b>Eye/Face Protection</b>    | Safety glasses with side-shields<br>or<br>Goggles                                                                                                                                            |        |        |                      |
| <b>Hand protection</b>        | Chemical-resistant gloves made of waterproof material such as neoprene, butyl rubber, barrier laminate or nitrile rubber.                                                                    |        |        |                      |
| <b>Body Protection</b>        | Wear long-sleeved shirt and long pants and shoes plus socks.                                                                                                                                 |        |        |                      |
| <b>Respiratory protection</b> | In case of mist, spray or aerosol exposure wear suitable personal respiratory protection and protective suit.                                                                                |        |        |                      |
| <b>Exposure Limits</b>        |                                                                                                                                                                                              |        |        |                      |
| 1,2-Propanediol               | 57-55-6                                                                                                                                                                                      | WEEL   | TWA    | 10 mg/m <sup>3</sup> |
|                               |                                                                                                                                                                                              | TX ESL | ST ESL | 20 ug/m <sup>3</sup> |
|                               |                                                                                                                                                                                              | TX ESL | ST ESL | 500 ppm              |
|                               |                                                                                                                                                                                              | TX ESL | AN ESL | 2 ug/m <sup>3</sup>  |
|                               |                                                                                                                                                                                              | TX ESL | AN ESL | 50 ppm               |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                |                                 |
|--------------------------------|---------------------------------|
| <b>Appearance</b>              | brown                           |
| <b>Physical State</b>          | liquid                          |
| <b>Odor</b>                    | characteristic                  |
| <b>pH</b>                      | 5.5 - 7.5 (100 %)               |
| <b>Vapor Pressure</b>          | no data available               |
| <b>Vapor Density (Air = 1)</b> | no data available               |
| <b>Density</b>                 | 1.02 g/cm <sup>3</sup> at 20 °C |
| <b>Evaporation rate</b>        | no data available               |
| <b>Boiling Point</b>           | no data available               |

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|                                               |                   |
|-----------------------------------------------|-------------------|
| <b>Melting / Freezing Point</b>               | no data available |
| <b>Water solubility</b>                       | dispersible       |
| <b>Minimum Ignition Energy</b>                | no data available |
| <b>Partition coefficient: n-octanol/water</b> | no data available |
| <b>Viscosity</b>                              | 150 - 400 mPa.s   |

**SECTION 10. STABILITY AND REACTIVITY**

|                            |                                 |
|----------------------------|---------------------------------|
| <b>Conditions to avoid</b> | freezing                        |
| <b>Incompatibility</b>     | no data available               |
| <b>Chemical Stability</b>  | Stable under normal conditions. |

**SECTION 11. TOXICOLOGICAL INFORMATION**

Acute toxicity studies have been bridged from a similar formulation containing a higher percentage of the active ingredient, spinosad. The non-acute information pertains to technical-grade spinosad.

|                                  |                                                                                                                                                                                                                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Acute oral toxicity</b>       | female rat: LD50: > 5,000 mg/kg<br>(Up and Down Method)                                                                                                                                                                                                                                       |
| <b>Acute dermal toxicity</b>     | male/female combined rat: LD50: > 5,000 mg/kg                                                                                                                                                                                                                                                 |
| <b>Acute inhalation toxicity</b> | male/female combined rat: LC50: > 2.5 mg/l<br>Exposure time: 4 h<br>Determined in the form of liquid aerosol.<br>(actual)<br><br>male/female combined rat: LC50: > 10 mg/l<br>Exposure time: 1 h<br>Determined in the form of liquid aerosol.<br>Extrapolated from the 4 hr LC50.<br>(actual) |
| <b>Skin irritation</b>           | rabbit: Slight irritation                                                                                                                                                                                                                                                                     |
| <b>Eye irritation</b>            | rabbit: Mild eye irritation.                                                                                                                                                                                                                                                                  |
| <b>Sensitisation</b>             | guinea pig: Non-sensitizing.                                                                                                                                                                                                                                                                  |
| <b>Chronic toxicity</b>          | Spinosad caused effects in the liver, kidney, spleen and/or thyroids in chronic dietary studies in rats and dogs.                                                                                                                                                                             |

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**Assessment Carcinogenicity**

Spinosad was not carcinogenic in lifetime feeding studies in rats and mice.

**ACGIH**

None.

**NTP**

None.

**IARC**

None.

**OSHA**

None.

**Reproductive toxicity**

REPRODUCTION: Spinosad is not considered a reproductive toxicant at non-maternally toxic dose levels.

DEVELOPMENTAL TOXICITY: Spinosad was not a developmental toxicant in rats and rabbits.

**Assessment neurotoxicity**

Spinosad did not cause neurotoxicity in rats in acute, subchronic or chronic toxicity studies.

**Mutagenicity**

Spinosad was not mutagenic or genotoxic in a battery of in vitro and in vivo mutagenicity studies.

**SECTION 12. ECOLOGICAL INFORMATION****Environmental precautions**

Apply this product as specified on the label. Do not contaminate surface or ground water by cleaning equipment or disposal of wastes, including equipment wash water. Do not allow to get into surface water, drains and ground water.

**SECTION 13. DISPOSAL CONSIDERATIONS****General Disposal Guidance**

Dispose in accordance with all local, state/provincial and federal regulations. Pesticide, spray mixture or rinse water that cannot be used according to label instructions may be disposed of on site or at an approved waste disposal facility.

**Container Disposal**

Do not re-use empty containers. Triple rinse containers. Dispose of empty container in a sanitary landfill or by incineration, or, if allowed by State/Provincial and local authorities, by burning. If burned, stay out of smoke.

**RCRA Information**

Characterization and proper disposal of this material as a special or hazardous waste is dependent upon Federal, State and local laws and are the user's responsibility. RCRA classification may apply.

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**SECTION 14. TRANSPORT INFORMATION**

According to national and international transport regulations this material is not classified as dangerous goods / hazardous material.

Freight Classification: INSECTICIDES OR FUNGICIDES, N.O.I., OTHER THAN POISON

**SECTION 15. REGULATORY INFORMATION**

EPA Registration No. 264-995

**US Federal Regulations****TSCA list**

1,2-Propanediol 57-55-6

**US. Toxic Substances Control Act (TSCA) Section 12(b) Export Notification (40 CFR 707, Subpt D)**

None.

**SARA Title III - Section 302 - Notification and Information**

None.

**SARA Title III - Section 313 - Toxic Chemical Release Reporting**

None.

**US States Regulatory Reporting****CA Prop65**

This product does not contain any substances known to the State of California to cause cancer.

This product does not contain any substances known to the State of California to cause reproductive harm.

**US State Right-To-Know Ingredients**

1,2-Propanediol 57-55-6 MN, RI

**Canadian Regulations****Canadian Domestic Substance List**

1,2-Propanediol 57-55-6

**Environmental****CERCLA**

None.

**Clean Water Section 307 Priority Pollutants**

None.

**Safe Drinking Water Act Maximum Contaminant Levels**

None.



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### SECTION 16. OTHER INFORMATION

NFPA 704 (National Fire Protection Association):

Health - 1

Flammability - 2

Instability - 0

Others - none

HMIS (Hazardous Materials Identification System, based on the Third Edition Ratings Guide)

Health - 1

Flammability - 1

Physical Hazard - 0

PPE -

0 = minimal hazard, 1 = slight hazard, 2 = moderate hazard, 3 = severe hazard, 4 = extreme hazard

Reason for Revision: Reviewed and updated for general editorial purposes. The following sections have been revised: Section 9: Physical and Chemical Properties. Section 16: Other Information.

Revision Date: 02/22/2012

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